

UNIT - 1

A) BASICS

- Cryptography : Introduction
- Security Goals :
 - Confidentiality
 - Integrity
 - Availability
- Common Threats :
 - Threat to Confidentiality :
 - Snooping
 - Traffic Analysis
 - Threat to Integrity :
 - Modification
 - Spoofing (Masquerading)
 - Replaying Attack
 - Repudiation
 - Threat to Availability :
 - Denial of Service (DoS)
- Passive and Active Attacks
- Security Services and Security Mechanisms
- Techniques : Cryptography and Steganography

B) MATHS AND MORE

- Mathematics of Cryptography :
 - Integer Arithmetic
 - Binary Operations
 - Greatest Common Divisor (GCD)
 - Euclidean Algorithm (EA)
 - Extended EA
 - Modular Arithmetic
 - Congruence
- Traditional Symmetric Key Ciphers :
 - Introduction
 - Kerckhoff's Principle
- Cipher-Text Only Attacks :
 - Techniques : Brute-Force, Statistical and Pattern
- Chosen-Plaintext Attack
- Chosen-Cipher Text Attack

C) TYPES OF CIPHERS

- Categories of Traditional Ciphers :
 - Substitution :
 - Mono-Alphabetic
 - Poly Alphabetic
 - Transposition
- Mono-Alphabetic :
 - Additive
 - Multiplicative
 - Affine

- Poly-Alphabetic :
 - Autokey
 - Playfair
 - Vigenere
 - Hill
 - One-Time Pad
 - Rotor
- Transposition (*only theory*) :
 - Keyless - Rail Fence Cipher
 - Keyed
- Stream and Block Ciphers (*only theory*)

UNIT - 2

- Modern Block Ciphers (MBC)
- Components of MBC :
 - P-Block (*Permutation Box*) :
 - Straight
 - Compression
 - Expansion
 - S-Block (*Substitution Box*) :
- Circular Shift
- Product Ciphers :
 - Diffusion
 - Confusion
 - Classes :
 - Feistel Ciphers
 - Non-Feistel Ciphers
- Feistel Ciphers :
 - What ?
 - Working
 - Example - **DES**
 - Final Design (Diagram)
- Non-Feister Ciphers :
 - What ?
 - Working
 - Example - **AES**
 - Final Design (Diagram)

- DES Function

Diagram and Algorithm -> Very Important
Any 1 of these might come as a single 4
Marker OR
All might be asked but without any points

- Expansion P-Box
- Whitener (XOR)
- S-Boxes
- Cipher and Reverse Cipher
- Complete Diagram
- Algorithm with 4 corresponding routines
- Properties :
 - Avalanche Effect
 - Completeness Effect
- Weaknesses :
 - S-Box : 3 and P-Box : 2
 - Cipher Key
- Modes of Operations of MBC (*diagrams*):
 - Electronic Codebook (ECB)
 - Cipher Block Chaining (CBC)
 - Cipher Feedback (CFB)
 - Output Feedback (OFB)
 - Counter (CTR)
- Use of Stream Ciphers :
 - RC4 :
 - State
 - Idea
 - Initialization
 - Key Stream Generation
 - Security Issues
 - One-Time Pad